Pipelining

Consider you are reading an article from English Literature. You ~~once~~ read a word "reckon" and you wish to know the meaning. For that you will Lookup the meaning of "reckon" in the Dictionary. The steps ~~included~~ include;

1. Fetch the word in the dictionary
2. Dictionary will Decode the word
3. You will Execute the sentence (understand)

Computer also does the same process.

Fetch ____ Decode ____ Execute

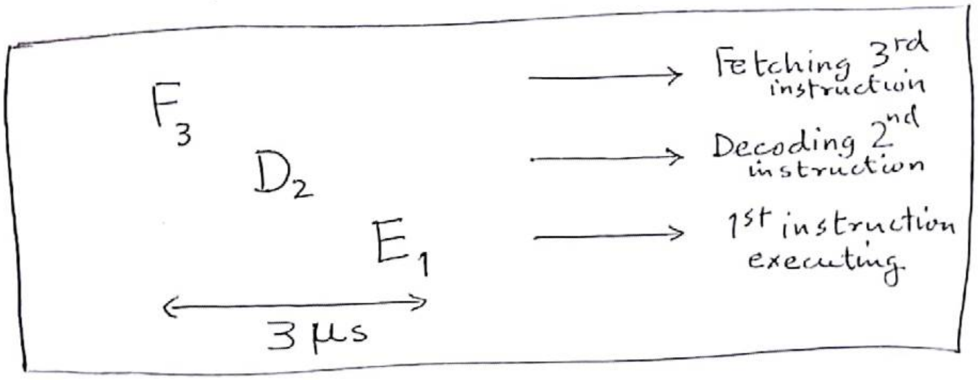
for any given instruction.

Consider if each component i.e.; Fetch, Decode, Execute takes 1 μs to complete, then the whole process will take 3 μs to complete.

In Microprocessor, the heart of a computer, there are two blocks for the purpose of Fetching, Decoding, and Exeuting the instructions. These blocks are named as Bus Interface Unit (BIU) and Execution Unit (EU).

Bus Interface Unit has the job to Fetch and Decode while 'EU' has the job to Execute the instructions. Obviously, not all the instruction are meant to be processed at the same times, i.e., they are executed one by one. These instructions, following the above timings may take 3 μs per instruction, making the next instruction to wait 3 μs.

What if we divide the units (BIU and EU) and provide a temporary storage to store a limited amount of instructions so that if one instructed ~~in~~ is Fetched it can be passed to the decoder part and the ~~other~~ second instruction can be fetched. as depicted in the figure below;



We can say that 3 instructions are in Pipeline to be executed in the sametime it took one instruction to get executed.

BIU stores the next fetched and decoded instructions in a FIFO memory that is Queue. Such that it canbe passed to the EU when the current instruction get exeuted.

The size of Queue for 8086 is 6 bytes and for 8088 is 4 bytes.

BIU has all the Segment Registers and ~~IP, BP~~, index and pointer Registers to support address calculation. Segment regislers include CS, DS, ES, SS. Index and Pointer registers include IP, ~~BP, SP, SI, DI~~

EU has all the general purpose regiters AX, BX, CX, DX and Flags. The main component of EU is Arithmetic Logic Unit (ALU).

BP, SP, SI, and DI are included in EU.

Since, there is a limited size of Queue, it poses a disadvantage ~~when~~ ~~a~~ for following cases:

Case I — Data for execution is not in memory

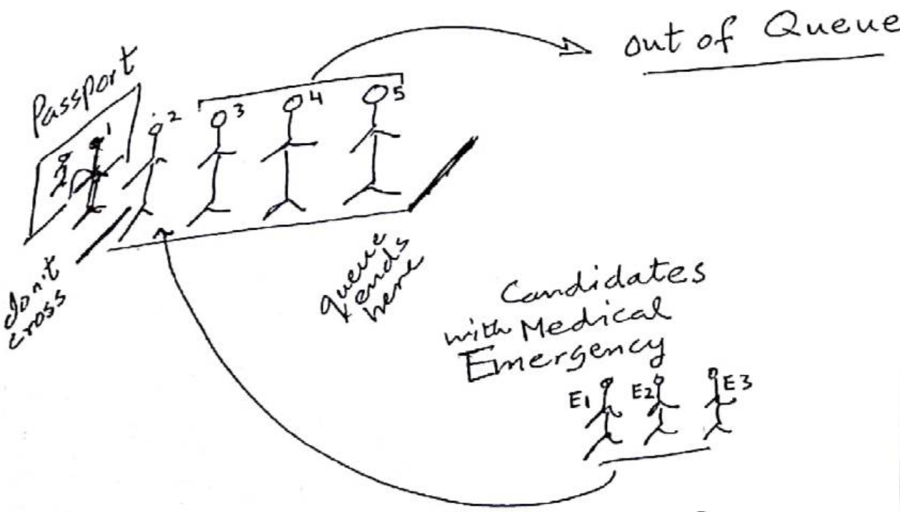
case II — Instruction is a Branch Instruction.

Case I: Data have to be read into the memory and the ~~in~~ subsequent instructions may wait for their turn for execution.

Case II: Branch Instructions are instructions which affect the flow such as "If-then" instructions or "For" "While" "do-while" i.e.; Loop instructions.

In these ~~cases~~ ("if-then" or "Loop") cases the ~~all~~ already fetched and decoded instruction should be discarded to load the instructions of 'if-then' or 'Loop' scenario.

Real Life example:



At window 1 person takes 5 minutes. Its already 4:35 pm and Window will be closed at 5:00 pm. This suggests 3, 4, 5. tokens will be discarded to accommodate E1, E2, E3.

This is called "Branch Penalty".